



BITS Pilani presentation

BITS Pilani
Pilani Campus

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EEE ZG512 / SS ZG512, EMBEDDED SYSTEM DESIGN

Lecture No. 02

Challenges in Embedded Systems Design



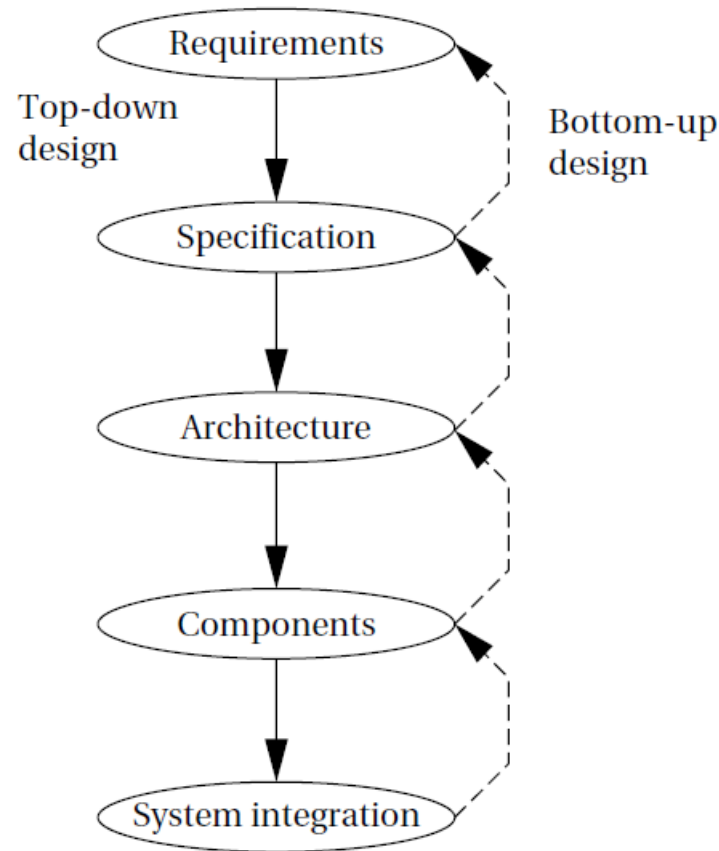
- Hardware requirement
- Deadline satisfaction
- Power optimization
- Complex testing
- Limited observability and controllability
- Restricted development environment
- Upgradability
- Operation & Maintenance

Performance in ES Computing



- CPU: Influences on behavior of system.
- Platform: I/O components affect the performance.
- Program: Structure of program.
- Task: Multitasking system.
- Multiprocessor: Complexity in Inter Processor Communication (IPC).

Abstract Layers in Design Process



Requirements:



- Performance
- Cost
- Physical size & weight
- Power consumption

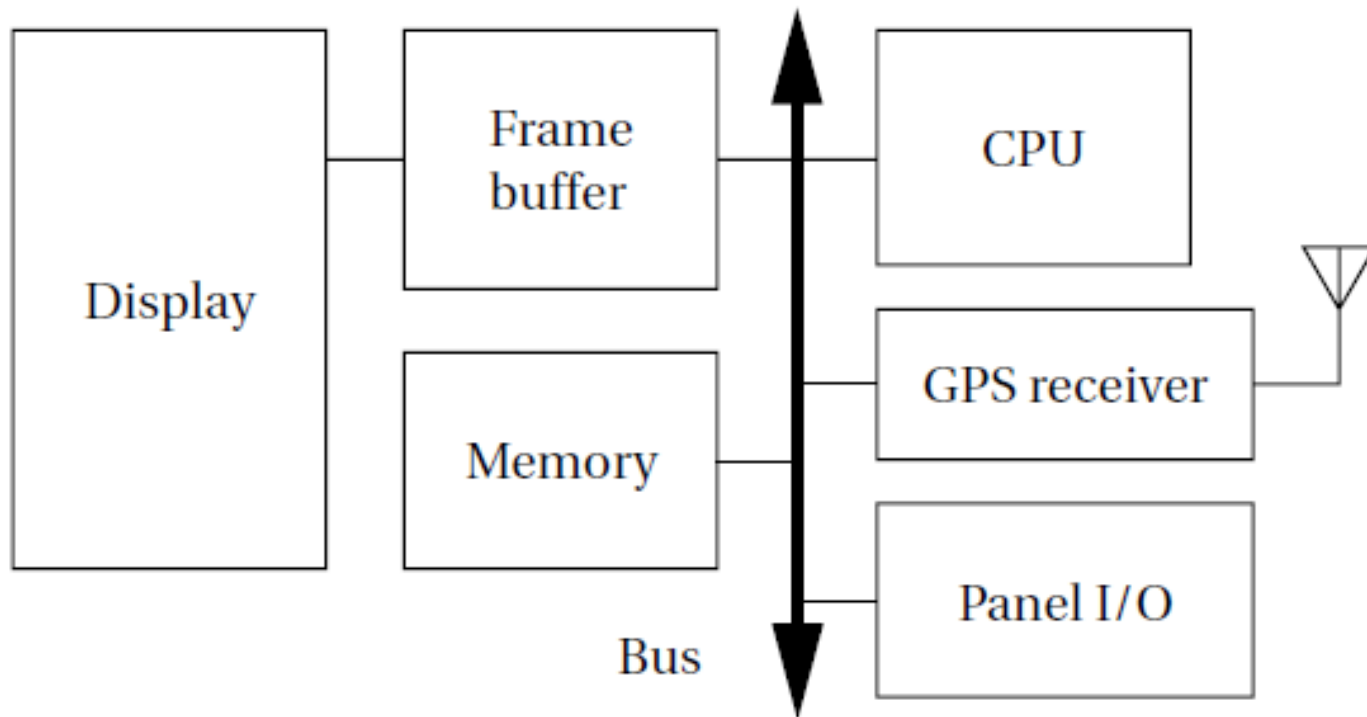


Formalism of System Design



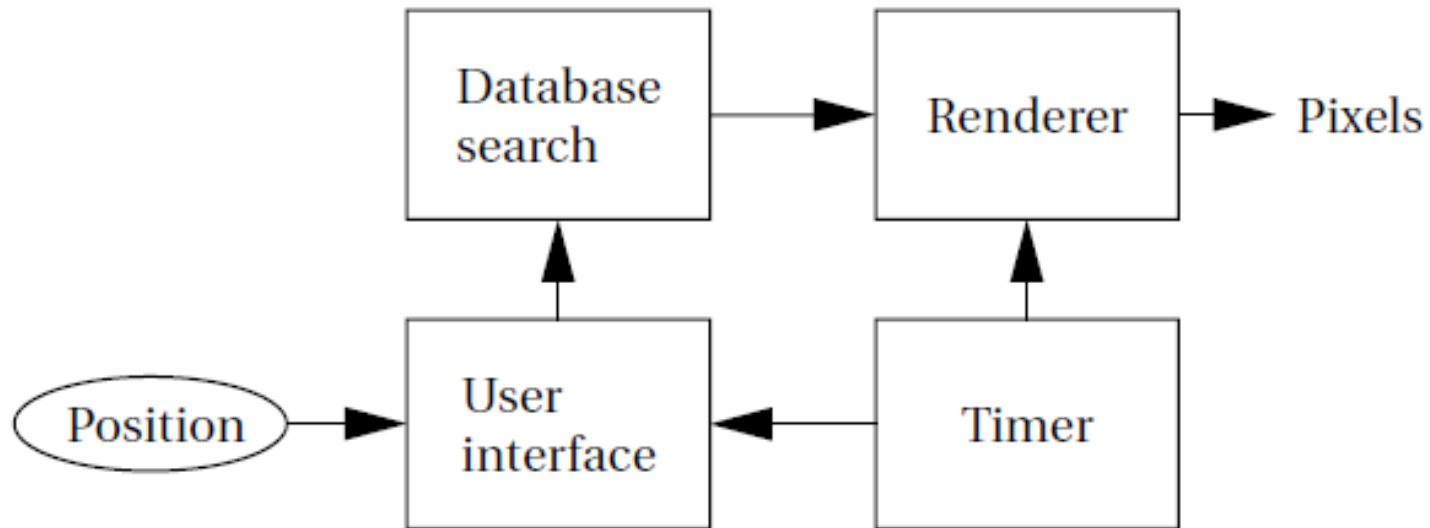
- Unified Modeling Language
- Structural description
- Behavioral description
 - Signal
 - Call event
 - Timeout event

Design Architecture



Hardware

Design Architecture



Software



Classification



- Complexity:
 - Small scale: 8/16-bit Microcontrollers
 - Medium: 16/32-bit Microcontrollers
 - Sophisticated: 64-bit Microcontrollers
- Operation:
 - Real time
 - Non-real time

Classification Continued...



- Time constraint:
 - Hard: Stringent deadline
 - Soft: Flexible deadline
 - Firm: Flexible deadline
- Implementation:
 - Distributed / Networked
 - Stand Alone

END
of
Session – 2
Thank You

